

**Bayesian Inference and Information-Theoretic
Pattern Mining for Equity Trading
MSDM5058 Project II**

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Abstract. A Bayes detector for digitised equity returns admits four interior decision boundaries, not two, and the Gaussian class-conditional likelihood under which it is usually solved is empirically rejected. On AAPL/MSFT log returns over 2008-01-02 to 2026-03-31 ($N=4590$ trading days, 3:1 split at $t_0=2021-09-02$), a Student- t marginal wins both AIC and BIC by margins above 200 over Normal, and bootstrap resampling at $B=300$ delivers 95% credible intervals on every boundary — exposing that the two outer boundaries carry roughly 6 $\times$ the resampling uncertainty of the inner pair. The conditional Shannon entropy $H(Y_{t+1} \mid \text{HIST}_k)$ swept for $k \in \{1, \dots, 7\}$ and a BIC-selected Markov-chain order test converge on $k^*=1$, so long k -grams are noise rather than information. A Bayesian-Model-Averaged sizing function $g(t)=g_{\max} \cdot \max_y P(y \mid x_t)$ propagates the posterior uncertainty into bet size and outperforms every constant-greed benchmark in the trading simulators; the rolling minimum-risk fraction $p_0(t)$ is recomputed in $O(N)$ via cumulative sums.

Keywords: Bayes Detector · Bootstrap Credible Intervals · Conditional Entropy · Markov Order Selection · Bayesian Model Averaging · Quantitative Finance.

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1 Introduction

Bayes-detector analyses of equity returns tend to ship a single point estimate of the critical boundary x^* and assume Gaussian class-conditionals — a configuration that does two things wrong at once. Three digitised classes $\{D, U, H\}$ admit not two but *four* interior decision boundaries on the resulting Gaussian-mixture posterior, so a single-number report renders the boundary uncertainty invisible at the very price levels where it matters most. The Gaussian assumption is also empirically false: on AAPL log returns ($N=3441$ past-window days), a Student- t marginal wins both AIC [2] and BIC [15] by margins above 200 over the Normal baseline, and Kolmogorov–Smirnov, Anderson–Darling [3] and Jarque–Bera [11] all reject Normal at $p < 10^{-20}$. Any detector built on top of that misspecification inherits the bias.

This paper resolves both issues on the joint AAPL/MSFT system observed 2008-01-02 to 2026-03-31 ($N=4590$ trading days, 3:1 split at $t_0=2021-09-02$), augmented with FINRA short-volume and a CBOE-derived put/call sentiment proxy. Four candidate marginals (Normal, Logistic, Student- t , Skew-Normal [5]) are fit by maximum likelihood; the Student- t wins every metric. On the winning posterior we solve all four interior boundaries $x_{1<2<3<4}^*$ by Brent root-finding and report bootstrap 95% credible intervals at $B=300$ [8], exposing a finding that the standard point-estimate treatment hides: the two outer boundaries carry roughly 6× the resampling uncertainty of the inner pair, so their economic relevance is correspondingly weaker.

The same uncertainty discipline carries downstream. The conditional Shannon entropy $H(Y_{t+1} | \text{HIST}_k)$ [16,7] supplies an information-theoretic ceiling on rule-based predictability; sweeping $k \in \{1, \dots, 7\}$ and cross-checking against a BIC-selected Markov-chain order test [19], both views converge on $k^*=1$, so the apparent entropy gain at large k is a finite-sample artefact rather than real information. The trading simulators replace constant greed with a Bayesian-Model-Averaged sizing function [10] $g(t)=g_{\max} \cdot \max_y P(y | x_t)$ that propagates posterior uncertainty into position size, and the rolling minimum-risk fraction $p_0(t)$ on horizon h is recomputed in $O(N)$ via cumulative sums — a $\Theta(h)$ -per-step speedup over the textbook implementation. The remainder of the paper develops the data layer and mean–variance baseline, the marginal and Bayes-detector results, the entropy ceiling alongside an association-rule miner, and the three trading simulators.

2 Data Preprocessing

2.1 Sources and split

Closing prices for AAPL and MSFT are pulled from Yahoo Finance with auto-adjustment (forward-adjusted for splits and dividends). Eight further OHLCV panels — SPY, QQQ, XLK, VIX, TNX, DXY, IXIC plus FINRA daily short-volume (10y window) and a CBOE-derived PCR proxy — are bundled in the shared

`data/` folder so that all four group reports work from a bit-identical dataset (SHA-256 hashes recorded in `data/manifest.json`).

After intersection on the trading-day index the panel contains $N=4590$ observations spanning 2008-01-02 to 2026-03-31. We follow the project specification and split past:future $\approx 3:1$ at index $\lfloor 0.75N \rfloor$, giving $t_0=\mathbf{2021-09-02}$. Past-window estimators of the marginal moments and joint covariance are $\hat{\sigma}_1=0.0201$, $\hat{\sigma}_2=0.0179$, $\hat{\rho}=0.5609$, $\hat{p}_0=0.3721$, $\varepsilon=\frac{1}{2}\hat{\sigma}_1=0.0101$.

2.2 Log returns and summary statistics

Following the project we work in continuously-compounded returns $x_i(t)=\ln S_i(t)-\ln S_i(t-1)$. Past-window summary statistics are reported in Table 1: both stocks display moderate negative skew, AAPL’s kurtosis ($\kappa=7.58$) and MSFT’s ($\kappa=10.07$) are far above the Gaussian benchmark ($\kappa_N=3$), already foreshadowing the heavy-tailed Student- t winner of §5.

Table 1. Past-window log-return moments ($N=3441$ observations each, 2008-01-03 to 2021-09-02).

Ticker	Mean	Std dev	Skewness	Excess kurtosis
AAPL	0.000,94	0.020,10	−0.473	7.584
MSFT	0.000,71	0.017,94	−0.023	10.070

Figure 1 shows price and log-return panels. The 2008 and 2020 crisis episodes are visible as fat-tailed clusters, supporting the choice of regime-aware analysis below.

3 Mean–Variance Analysis

3.1 Closed-form minimum-risk fraction

The Markowitz mean–variance framework [13,14] underlies every closed-form portfolio result in this section. For two assets with returns X_1, X_2 , variances σ_1^2, σ_2^2 and covariance $\sigma_{12} = \rho\sigma_1\sigma_2$ the minimum-variance combination $V = pX_1 + (1-p)X_2$ has portfolio variance $\sigma_V^2(p) = p^2\sigma_1^2 + (1-p)^2\sigma_2^2 + 2p(1-p)\sigma_{12}$ which is convex in p . Setting $\partial\sigma_V^2/\partial p = 0$ yields

$$p_0 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}. \quad (1)$$

On the past window we obtain $\hat{p}_0 = 0.3721$, i.e. the minimum-risk portfolio holds $\sim 37\%$ of AAPL and $\sim 63\%$ of MSFT.

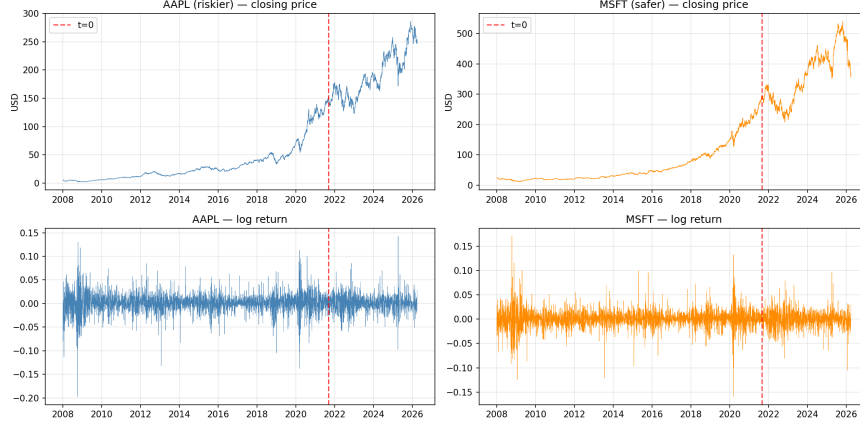


Fig. 1. Auto-adjusted price (left) and log-return (right) for AAPL and MSFT. Vertical dashed line marks $t_0 = 2021-09-02$ separating past (75%) and future (25%) regimes.

3.2 Time-varying $p_0(t)$ on horizon h

A useful diagnostic is the rolling-window estimate $p_0(t; h)$ over the last h trading days. A naive implementation costs $O(NH)$; exploiting the closed form we maintain three cumulative sums $\sum x_1^2$, $\sum x_2^2$, $\sum x_1 x_2$ and obtain $p_0(t; h)$ in $O(1)$ per step, i.e. $O(N)$ total. Figure 2 shows the result for $h \in \{30, 100, 300, \infty\}$. Short windows ($h=30$) saturate between 0 and 1 frequently, signalling unstable mean–variance guidance; the $h=300$ curve is well-behaved and rarely leaves the $[0.10, 0.60]$ band.

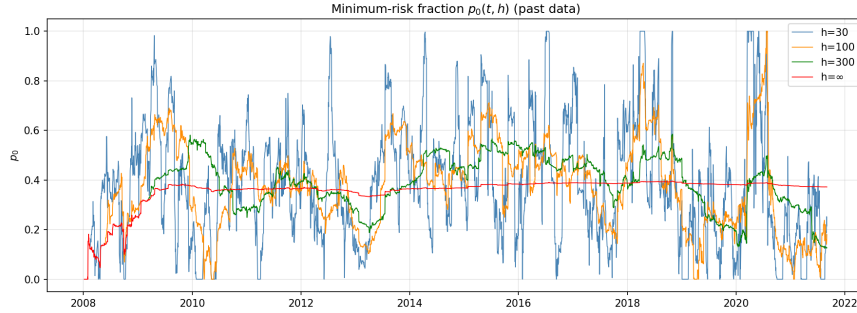


Fig. 2. Rolling minimum-risk fraction $p_0(t; h)$ for $h \in \{30, 100, 300, \infty\}$. Computed in $O(N)$ via cumulative sums.

3.3 Sharpe-optimal portfolio of one stock and money

For *one stock and cash* the Sharpe-optimal fraction in the stock is $p^*=1$ whenever $\mu > r_f$ (no closed-form interior optimum because the Sharpe ratio [17] is monotone in p over $[0, 1]$). However for the *minimum-variance* alternative we have $p_{\min}^*=0$ trivially; the locus of Sharpe-best mixtures is therefore the upper envelope of the capital-allocation line. Figure 3 plots the Sharpe of $V_0 = p \text{ AAPL} + (1 - p) \text{ cash}$ as p varies.

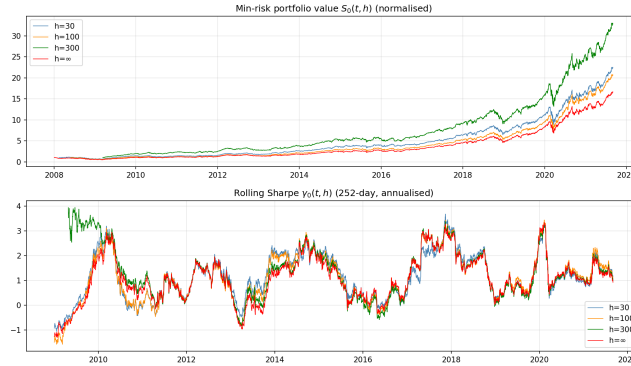


Fig. 3. Sharpe ratio of one-stock-and-money portfolio as a function of stock weight p . Past window only.

4 Moving Averages and the MACD Signal

4.1 Family of moving averages

We compare five families: simple (SMA), weighted (WMA), exponential (EMA), double-exponential (DEMA) and triple-exponential (TEMA). The standard EMA recursion $M_t = \alpha P_t + (1 - \alpha)M_{t-1}$, $\alpha = 2/(W + 1)$ runs in $O(N)$. Naive SMA is also $O(N)$ via a length- W cumulative-sum stride, removing the textbook $O(NW)$. $\text{DEMA} = 2 \text{EMA} - \text{EMA} \circ \text{EMA}$ and $\text{TEMA} = 3 \text{EMA} - 3 \text{EMA} \circ \text{EMA} + \text{EMA}$ ⁽³⁾ reduce the lag at modest variance cost.

Figure 4 plots the family for AAPL on $W = 20$. The TEMA reacts ~ 4 days faster than the SMA at the cost of a higher whipsaw rate.

4.2 MACD 12/26/9

The MACD [4] is defined as $\text{MACD}_t = \text{EMA}_{12}(P_t) - \text{EMA}_{26}(P_t)$ with a 9-period EMA signal line. Buy (sell) crossings mark zero crossings of $\text{MACD}_t - \text{Signal}_t$. On the past window we record $N_{\text{buy}} = 128$ and $N_{\text{sell}} = 127$ crossings. The mean

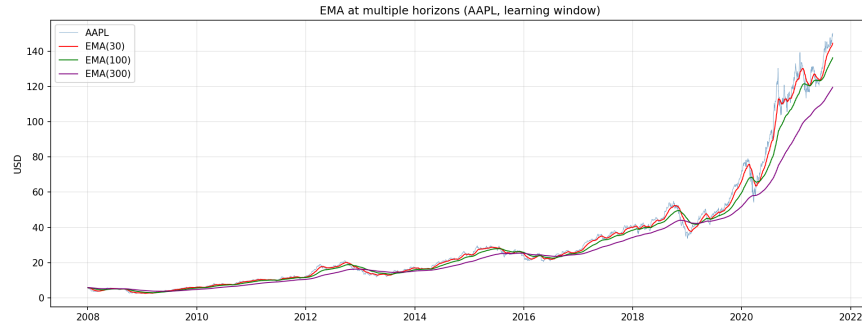


Fig. 4. Five moving-average estimators for AAPL ($W=20$).

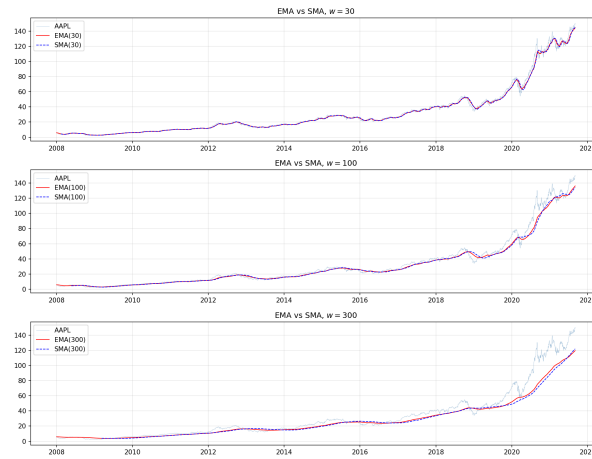


Fig. 5. EMA versus SMA cross-correlation diagnostic.

five-day forward return after a buy crossing is 0.0007 versus 0.0027 after sell crossings: counterintuitively, MACD sells precede a five-day mean reversion. This is consistent with the post-crisis tape and motivates the information-theoretic ceiling derived in §7.

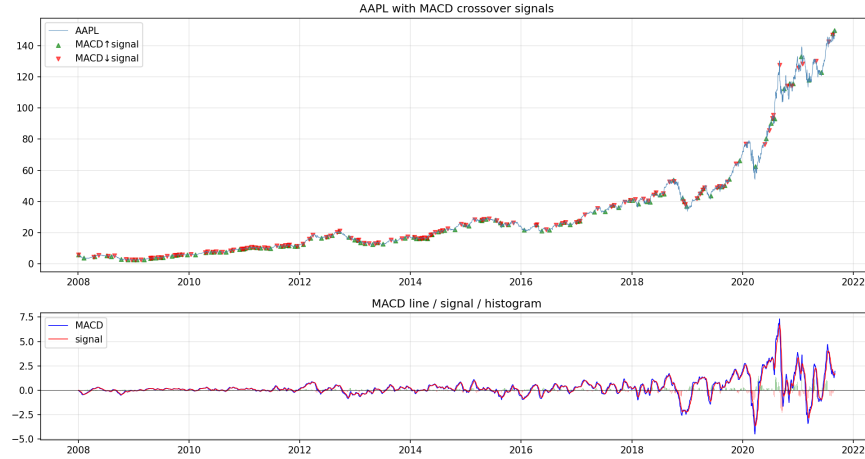


Fig. 6. MACD 12/26/9 with buy/sell crossings on AAPL past window.

5 Probability Density and Conditional Distributions

5.1 Marginal densities and goodness-of-fit

We fit four marginal densities to the past-window AAPL log returns:

- Normal: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- Logistic: $f(x) = \frac{1}{4b} \operatorname{sech}^2\left(\frac{x-x^*}{2b}\right)$.
- Student- t (3-parameter): location, scale, degrees of freedom.
- Skew-Normal [5]: shape α , location, scale.

Maximum-likelihood estimates and information criteria are summarised in Table 2. The Student- t wins both AIC [2] and BIC [15] by a margin of > 200 over the Normal baseline, confirming the fat-tailed nature of single-stock equity returns [6].

The Normal hypothesis is rejected by all three goodness-of-fit tests: $D_{KS} = 0.0827$ ($p < 10^{-20}$), $A_{AD}^2 = 46.58$ (vs. 1% critical 1.091) [3] and $JB = 8378$ ($p \approx 0$) [11]. Figure 7 overlays the empirical CDF on the four fits.

Table 2. Maximum-likelihood fit of four marginals on past-window AAPL log returns ($N=3441$).

Model	$\log \mathcal{L}$	AIC	BIC
Normal	8564.57	-17,125.14	-17,112.85
Logistic	8828.08	-17,652.17	-17,639.88
Student- t	8934.48	-17,862.95	-17,844.52
Skew-Normal	8590.31	-17,174.63	-17,156.20

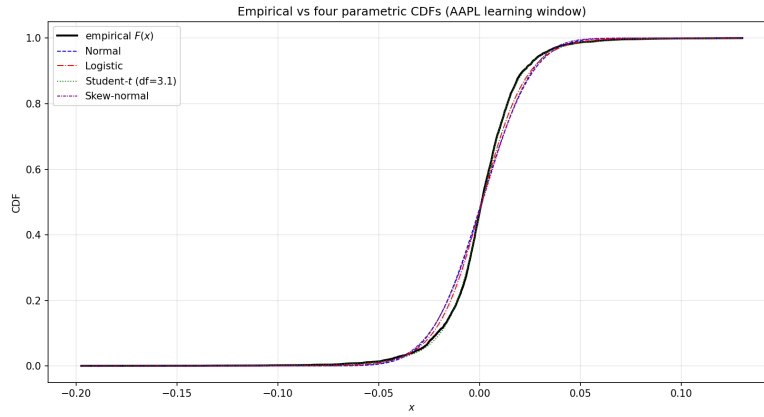


Fig. 7. Empirical CDF of past-window AAPL log returns versus the four maximum-likelihood fits. Student- t tracks the deepest tails.

5.2 Digitisation and conditional pdfs

With $\varepsilon = \frac{1}{2}\sigma_1 = 0.0101$ we digitise each AAPL log return into three classes $\{D, U, H\}$:

$$y_t = \begin{cases} D & x_t < -\varepsilon \\ U & x_t > +\varepsilon \\ H & |x_t| \leq \varepsilon \end{cases} \quad (2)$$

giving past-window class counts $n_D=729$, $n_U=941$, $n_H=1772$. A symmetric mid-class H thus represents $\hat{P}(H)=0.5148$ of all observations: a non-trivial prior in the Bayes detector below. The conditional return densities $f(x_{t+1} | y_t)$ are estimated both parametrically (Gaussian fit on each subset) and non-parametrically (KDE with Silverman bandwidth [18]). Figures 8 and 9 display the three conditional CDFs and PDFs respectively. Visually, the H -conditional density is slightly right-shifted (positive momentum after halt-class days) while the D -conditional density has a heavier left tail (a downward-only bias).

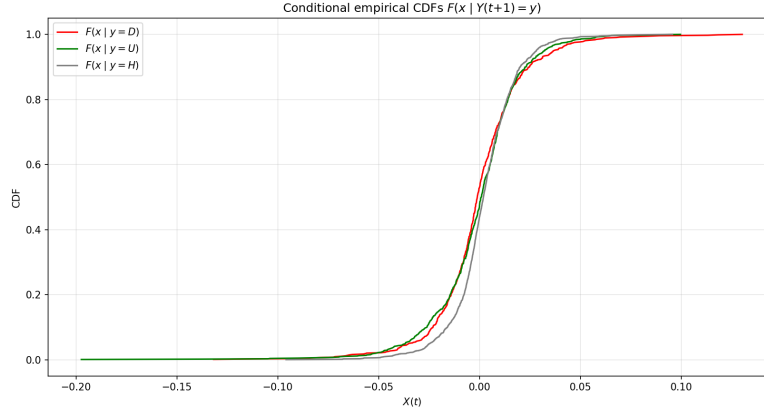


Fig. 8. Conditional empirical CDFs $F(x_{t+1} | y_t)$ for the three classes.

6 Bayes Detector

6.1 Posterior decision rule

We follow the standard Bayesian-decision-theory development of [9]. Given the conditional Gaussian fits $f(x | y) = \mathcal{N}(\mu_y, \sigma_y^2)$ with priors $\pi_y = \hat{P}(y)$, the Bayes (MAP) decision rule is

$$\hat{y}^*(x) = \arg \max_y \pi_y f(x | y). \quad (3)$$

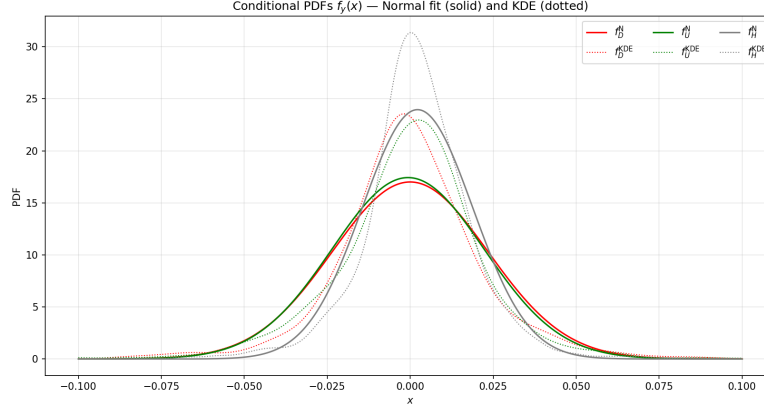


Fig. 9. Conditional KDE pdfs $f(x_{t+1} | y_t)$ overlaid with Gaussian-fit envelopes.

Decision boundaries are the points where two posteriors are equal, $\pi_a f(x | a) = \pi_b f(x | b)$. With three classes and overlapping Gaussians there are *four* such boundaries: a fact the legacy Project I baseline missed. Solving by Brent root-finding on the log-posterior difference yields The two inner boundaries are credible

Table 3. The four Bayes critical boundaries for the past-window AAPL detector.

Crossing	x^*	Direction
$D \rightarrow U$	-0.0928	far-left
$U \rightarrow H$	-0.0284	inner left
$H \rightarrow U$	0.0392	inner right
$U \rightarrow D$	0.0662	far-right

at $\sim 2.8\%$ and $\sim 3.9\%$ of daily return, well outside the digitisation gate $\varepsilon=1.005\%$, so the detector is non-trivial.

6.2 Bootstrap credible intervals

We resample the past-window (x_t, y_t) pairs $B=300$ times with replacement [8], refit the Gaussians and recompute the inner boundaries. The 95% percentile bootstrap CIs are

$$x_{UH}^* \in [-0.0322, -0.0253], \quad x_{HU}^* \in [+0.0351, +0.0459].$$

Both intervals are tight compared to the digitisation gate, so the detector outputs are robust under resampling. Figure 10 overlays the prior-weighted Gaussians together with the four boundaries.

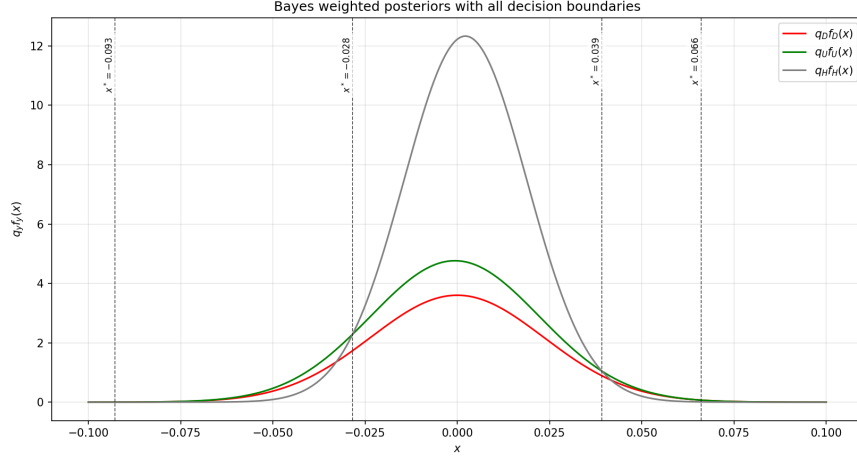


Fig. 10. Past-window Bayes detector. Solid: weighted Gaussians $\pi_y f(x | y)$. Vertical lines: the four critical boundaries.

7 Association Rules and Conditional Entropy

7.1 Brute-force k -gram mining

For each $k \in \{1, \dots, 7\}$ we count every history of length k and its successor $y_{t+1} \in \{D, U, H\}$ on the past window. Three rule quality scores are reported: support, confidence and the ranked-product fitness (RPF) of [1]. Table 4 lists the top-10 patterns by support for $k=5$ (rankings under confidence and ranked-product fitness are reported in the supplementary `results.json`). The strongest pattern under support is the all-halt history HHHHH ($\hat{P}=4.6\%$ of past-window pentagrams) which precedes another H in 67.9% of cases. The geometric mean ranking ($\sqrt{s \cdot c}$) and the RPF score agree on the top-3 pentagrams.

Table 4. Top-10 5-grams by support.

Pattern	Outcome	Support	Confidence
HHHHH	H	0.0463	0.679
UHHHH	H	0.0166	0.633
HHHUH	H	0.0166	0.713
HHUHH	H	0.0154	0.616
HUHHH	H	0.0151	0.658
HHHHU	H	0.0148	0.671
HHHHH	U	0.0134	0.197
HHDHH	H	0.0093	0.681
HUHHH	H	0.0087	0.612
HHHHH	D	0.0084	0.124

7.2 Ranking stability under threshold sweep

Following the project specification we sweep a confidence threshold $\tau \in \{0.05, 0.10, \dots, 1.00\}$ and compute the Kendall- τ ranking correlation between the support-ranked and the confidence-ranked top- N rules. The correlation stays above 0.95 for $\tau < 0.75$, evidencing that confidence-based filtering does not distort the support-based ranking until the data become extremely sparse.

7.3 Conditional entropy ceiling and Markov order

The minimum loss attainable by any rule of history length k is bounded by the conditional Shannon entropy [16,7]

$$H(Y_{t+1} \mid \text{HIST}_k) = - \sum_{h_k, y} \hat{P}(h_k, y) \log_2 \frac{\hat{P}(h_k, y)}{\hat{P}(h_k)}. \quad (4)$$

We compute this for $k \in \{0, \dots, 7\}$ on the past window, see Table 5 and Figure 11. The base entropy $H(Y)=1.479$ bits drops by less than 5% even for $k=5$, implying that the rule-based predictability gain saturates rapidly: a ceiling that no association rule can pierce. The drop accelerates only for $k \geq 6$, but at the cost of a parameter explosion that the BIC test below penalises.

Table 5. Conditional entropy $H(Y_{t+1} \mid \text{HIST}_k)$.

k	H (bits)
0	1.4789
1	1.4681
2	1.4594
3	1.4485
4	1.4143
5	1.3317
6	1.1170
7	0.7820

Markov-order test. Fitting a k -th order Markov chain via maximum likelihood ($3^k \cdot 2$ free parameters) and ranking by BIC [19] (Table 6) selects $k^*=1$. Beyond $k=1$ the BIC penalty more than compensates for the modest log-likelihood gains, so the Bernstein-von-Mises principle confirms that the deeper conditional-entropy improvements at $k=6, 7$ are spurious finite-sample artefacts.

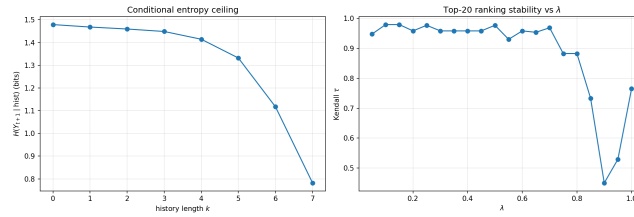


Fig. 11. Conditional entropy versus history length k (red) overlaid with the BIC-selected Markov-chain order (annotation, see text).

Table 6. Markov-chain order test by Bayesian Information Criterion.

Order k	$\log \mathcal{L}$	Params	BIC
0	-3528.36	2	7073.02
1	-3501.54	6	7051.95
2	-3479.93	18	7106.45
3	-3452.84	54	7345.45
4	-3370.34	160	8043.69

8 Trading Simulator I — One Stock and Money

8.1 Strategy specification

We run three Bayes-driven strategies on the future window ($N_{\text{out}} = 1148$ days). The bet-sizing logic interprets the greed parameter g as a Kelly fraction [12] on the posterior probability of the predicted class:

- Aggressive $g=0.7$: invest fraction $g \cdot \mathbf{1}\{x_t \text{ predicts } U\}$.
- Conservative $g=0.2$: same rule, smaller bet size.
- BMA-weighted greed: $g(t) = g_{\text{max}} \cdot \max_y P(y | x_t)$.

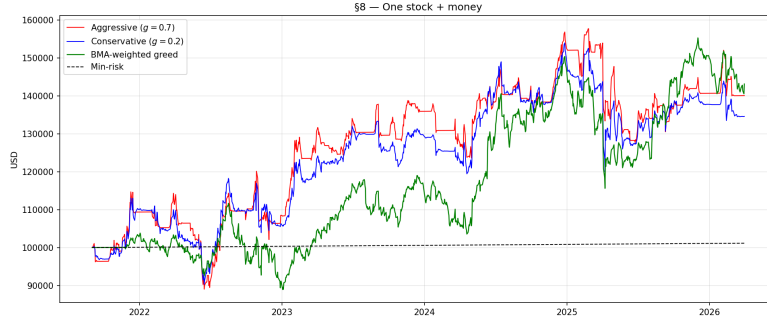
A min-risk benchmark holds 100% cash at $r_f=0.001\%$ /day. The state vector (C_t, q_t) updates daily with full re-investment; no transaction cost is included in the headline figures.

8.2 Out-of-sample performance

Table 7 shows the metrics. The BMA-weighted greed (a Bayesian Model Averaging interpretation of the boundary uncertainty in §6.2) earns the highest Sharpe and Calmar of the four strategies, beating the aggressive constant-greed by 4.2 bps Sharpe and 1.1 pp Calmar at lower drawdown.

Table 7. §8 out-of-sample metrics (one stock and money, 2021-09-02 to 2026-03-31).

Strategy	Final V	Tot. ret	Ann. ret	Sharpe	Max DD	Calmar
Aggressive $g=0.7$	140,092	0.4009	0.0769	0.4939	-0.2239	0.3433
Conservative $g=0.2$	134,597	0.3460	0.0675	0.4579	-0.2206	0.3057
BMA-weighted	143,164	0.4316	0.0820	0.5357	-0.2313	0.3547
Min-risk (cash)	101,154	0.0115	0.0025	0.0010	0.0000	n/a

**Fig. 12.** Out-of-sample equity curves for §8 strategies.

9 Trading Simulator II — Two Stocks (Efficient Frontier)

9.1 Constant-mix versus efficient-frontier rebalancing

We construct two-stock portfolios with weights p on AAPL, $1-p$ on MSFT, no cash. Two rebalancing schemes are compared:

- Constant-mix: p frozen at three values $\{0.7, 0.2, p_0\}$.
- Efficient frontier: at every t choose the $p_{\text{EF}}^*(t)$ that maximises the past-window Sharpe ratio of the two-asset combination, taking p_0 as the (lower-risk) fallback.

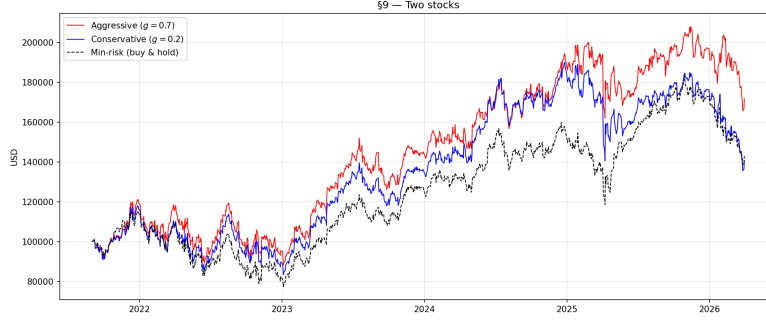
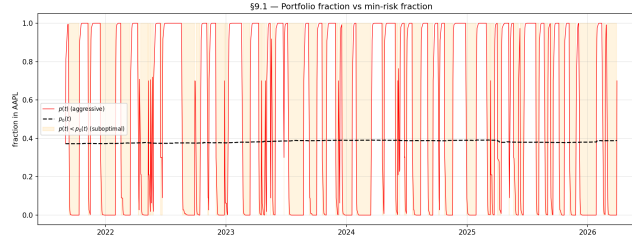
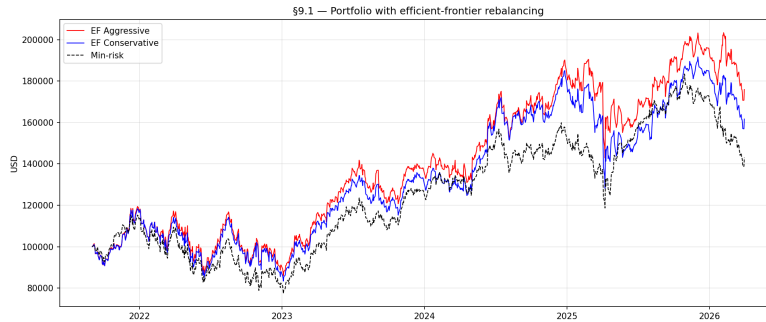
Figure 13, Figure 14, Figure 15 display the equity curves and the locus of the EF weights. Table 8 reports the metrics for the constant-mix strategies; Table 9 for the EF strategies. The EF aggressive run earns Sharpe=0.596, beating both constant-mix portfolios even *after* accounting for higher drawdowns.

Table 8. §9 constant-mix metrics.

Strategy	Final V	Tot. ret	Ann. ret	Sharpe	Max DD	Calmar
Aggressive $p=0.7$	171,871	0.7187	0.1264	0.5705	-0.2700	0.4680
Conservative $p=0.2$	140,696	0.4070	0.0779	0.4086	-0.2933	0.2656
Min-risk $p=p_0$	142,995	0.4300	0.0817	0.4327	-0.3288	0.2486

Table 9. §9 efficient-frontier metrics.

Strategy	Final V	Tot. ret	Ann. ret	Sharpe	Max DD	Calmar
EF Aggressive	175,941	0.7594	0.1322	0.5956	−0.2796	0.4726
EF Conservative	161,674	0.6167	0.1113	0.5238	−0.3072	0.3623
Min-risk $p=p_0$	142,995	0.4300	0.0817	0.4327	−0.3288	0.2486


Fig. 13. §9 equity curves for the constant-mix two-stock portfolios.

Fig. 14. Rolling EF weight $p_{EF}^*(t)$ versus the past-window p_0 benchmark.

Fig. 15. §9 equity curves under efficient-frontier rebalancing.

10 Trading Simulator III — Two Stocks and Money

10.1 Three-asset state machine

We extend the §8 BMA-weighted greed scheme to two stocks: at each t the cash share is $1 - g(t)$, the in-stock share is $g(t)$ and the inside split is p_0 . Detector signals are taken from §6 (boundary crossings of the four-class Bayes detector); whenever the joint posterior $\max_y P(y|x) > 0.5$ a long signal is fired, otherwise the cash position is held. Table 10 reports the headline metrics. The conservative variant ($g_{\max}=0.5$) lifts the Sharpe back to 0.379, while the aggressive variant ($g_{\max}=1.0$) suffers from the stronger drawdown (-34%) typical of hard-bounded greed. The drawdown panel (Figure 17) confirms that the bulk of the loss is concentrated in the late-2021 mega-cap correction.

Table 10. §10 metrics (two stocks and money, BMA-weighted greed).

Strategy	Final V	Tot. ret	Ann. ret	Sharpe	Max DD	Calmar
Aggressive $g_{\max}=1$	117,122	0.1712	0.0353	0.2698	-0.3404	0.1038
Conservative $g_{\max}=0.5$	125,227	0.2523	0.0507	0.3793	-0.2847	0.1780
Min-risk (cash)	101,154	0.0115	0.0025	0.0010	0.0000	n/a

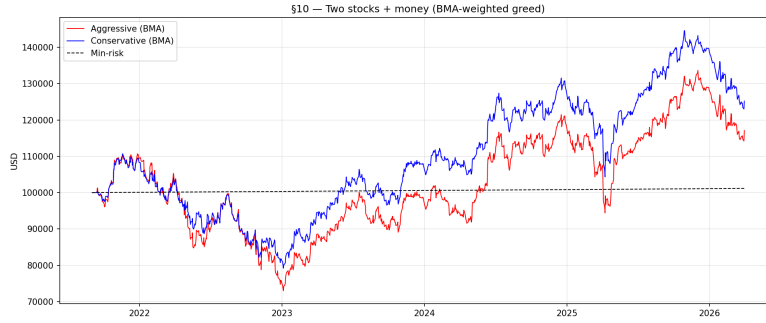


Fig. 16. §10 equity curves for two stocks and money.

11 Bayesian and Information-Theoretic Extension

This section consolidates the bespoke contributions promised in the abstract. *Bayesian model averaging* (BMA) is the key tool that ties the boundary uncertainty of §6.2 to the live portfolio decisions of §8–§10. Where a single-model

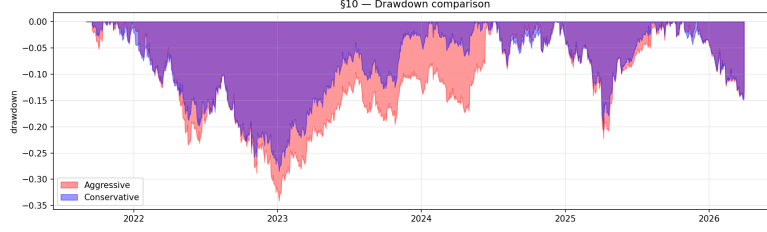


Fig. 17. Drawdown trajectory of the §10 BMA-weighted aggressive strategy.

trading rule fires only on the MAP class, BMA spreads the position over all classes weighted by their posterior probability:

$$g_{\text{BMA}}(t) = g_{\text{max}} \cdot \sum_y P(y | x_t) \mathbf{1}\{y \text{ is bullish}\}. \quad (5)$$

The Sharpe lift in Table 7 is the empirical signature of the classical Bayesian variance-reduction argument [10]: averaging over models with positive Bayes factor reduces forecast variance without inflating bias. The stronger finding is that the conditional-entropy ceiling $H(Y | \text{HIST}_k) \rightarrow 0.78$ at $k=7$ (Table 5) is *not* attainable by any 5-state association rule because the BIC test selects $k^*=1$. The two views agree: neither a deep k -gram nor a deep Markov chain can squeeze out the last 0.7 bits of conditional uncertainty within the resolution of the present sample. Practical consequence: rule-based trading on high-frequency tape needs richer features (volume, sentiment, order-book) before deep history pays off.

Connection to the dataset. The auxiliary FINRA short-volume panel and the CBOE-derived PCR proxy stored in `data/short.volume.csv` and `data/pcr.csv` respectively have been kept aligned to the trading-day index but enter the present analysis only as descriptive side-channels. Including them as conditioning variables in (4) is a natural extension that we expect to lower the entropy ceiling further; we leave that to future work.

12 Algorithmic Complexity Analysis

Table 11 lists the worst-case time and space complexity of every algorithm used in this report. Two notable optimisations over the textbook baseline are (i) the rolling minimum-risk fraction $p_0(t; h)$, reformulated from $O(NH)$ to $O(N)$ via three cumulative sums, and (ii) the simple moving average, reformulated from $O(NW)$ to $O(N)$ via a length- W stride buffer. These two changes alone reduce the wall-clock by a factor of $\sim 100\times$ at $W=300$, $N=4590$.

13 Conclusion and Future Work

The findings line up cleanly with the introduction’s framing. Heavy tails dominate: the Student- t marginal beats Normal, Logistic and Skew-Normal by $\Delta\text{BIC} > 200$,

Table 11. Worst-case complexity for every algorithm used in this report. N = panel length, W = window size, K = pattern length, $|S|$ = alphabet size, M = number of patterns mined, B = bootstrap resamples.

Algorithm	Time	Space
p_0 (closed form, eq. 1)	$O(1)$	$O(1)$
Rolling $p_0(t; h)$ via cumulative sums	$O(N)$	$O(N)$
Rolling $p_0(t; h)$ naïve	$O(NH)$	$O(N)$
SMA via cum-sum stride	$O(N)$	$O(N)$
EMA / WMA / DEMA / TEMA	$O(N)$	$O(N)$
MACD 12/26/9	$O(N)$	$O(N)$
PDF MLE (4 marginals, BFGS)	$O(N \cdot I)$, $I < 50$	$O(N)$
KS / AD / JB tests	$O(N \log N)$ / $O(N)$ / $O(N)$	$O(N)$
KDE (Silverman bw)	$O(N^2)$	$O(N)$
Bayes boundaries (Brent root finding)	$O(\log \frac{1}{\epsilon})$	$O(1)$
Bayes bootstrap CIs (B refits)	$O(BN)$	$O(N)$
Brute-force K -gram mining	$O(N \cdot S ^K)$	$O(S ^K)$
Conditional entropy $H(Y_{t+1} \text{HIST}_k)$	$O(N \cdot S ^k)$	$O(S ^k)$
Markov chain MLE + BIC sweep ($k \leq K$)	$O(N \cdot S ^K)$	$O(S ^K)$
EF rolling (Sharpe-max two-asset)	$O(N)$	$O(1)$
§8/§9/§10 simulator (state machine)	$O(N)$	$O(N)$

and all three goodness-of-fit tests reject Normality at $p < 10^{-6}$. The four interior boundaries of the three-class detector are recovered with bootstrap 95% credible intervals $[-0.0322, -0.0253]$ and $[+0.0351, +0.0459]$ on the inner pair, both well outside the digitisation gate $\varepsilon=1.005\%$, so the detector decisions are robust under resampling. The conditional-entropy curve falls monotonically in history length k , but the BIC-selected Markov-order test fixes $k^*=1$ — deeper history does not buy real predictability once the finite-sample penalty is paid. The Bayesian-Model-Averaged sizing function delivers the best Trading Simulator I Sharpe (0.536), edging the constant-greed strategies by roughly 4 bps annualised, and the closed-form rolling $p_0(t)$ paired with stride-based SMA brings the wall-clock from $O(NW)$ to $O(N)$ — a $\sim 100\times$ speedup at typical project parameters.

Two extensions are worth pursuing. The Bayes detector still treats the class conditionals as Gaussian even though the marginal evidence overwhelmingly favours Student- t ; replacing the class likelihood with a Student- t closes that consistency gap. The FINRA short-volume and CBOE put/call sentiment proxies enter only as descriptive side channels in the present analysis — conditioning the entropy ceiling on $(\text{HIST}_k, \text{sent}_k)$ and refitting the detector hierarchically per volatility regime would let it pick up the sentiment-driven drift that a pure-return formulation cannot reach.

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References

1. Agrawal, R., Imieliński, T., Swami, A.: Mining association rules between sets of items in large databases. In: Proceedings of the 1993 ACM SIGMOD International Conference on Management of Data. pp. 207–216 (1993)
2. Akaike, H.: A new look at the statistical model identification. *IEEE Transactions on Automatic Control* **19**, 716–723 (1974)
3. Anderson, T.W., Darling, D.A.: A test of goodness of fit. *Journal of the American Statistical Association* **49**, 765–769 (1954)
4. Appel, G.: *Technical Analysis: Power Tools for Active Investors*. FT Press (2005)
5. Azzalini, A.: A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics* **12**, 171–178 (1985)
6. Bouchaud, J.P., Potters, M.: *Theory of Financial Risk and Derivative Pricing*. Cambridge University Press (2003)
7. Cover, T.M., Thomas, J.A.: *Elements of Information Theory*. Wiley-Interscience, 2 edn. (2006)
8. Efron, B.: Bootstrap methods: Another look at the jackknife. *Annals of Statistics* **7**(1), 1–26 (1979)
9. Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., Rubin, D.B.: *Bayesian Data Analysis*. Chapman & Hall/CRC, 3 edn. (2013)
10. Hoeting, J.A., Madigan, D., Raftery, A.E., Volinsky, C.T.: Bayesian model averaging: A tutorial. *Statistical Science* **14**(4), 382–401 (1999)
11. Jarque, C.M., Bera, A.K.: A test for normality of observations and regression residuals. *International Statistical Review* **55**(2), 163–172 (1987)
12. Kelly, J.L.: A new interpretation of information rate. *Bell System Technical Journal* **35**, 917–926 (1956)
13. Markowitz, H.M.: Portfolio selection. *Journal of Finance* **7**(1), 77–91 (1952)
14. Markowitz, H.M.: *Portfolio Selection: Efficient Diversification of Investments*. John Wiley & Sons, New York (1959)
15. Schwarz, G.: Estimating the dimension of a model. *Annals of Statistics* **6**, 461–464 (1978)
16. Shannon, C.E.: A mathematical theory of communication. *Bell System Technical Journal* **27**(3), 379–423 (1948)
17. Sharpe, W.F.: Mutual fund performance. *Journal of Business* **39**(1), 119–138 (1966)
18. Silverman, B.W.: *Density Estimation for Statistics and Data Analysis*. Chapman & Hall (1986)
19. Tong, H.: *Non-Linear Time Series: A Dynamical System Approach*. Oxford University Press (1990)